

## DETAILED DESCRIPTION OF THE INVENTION — SENSOR SYSTEMS AND PHYSIOLOGIC MEASUREMENT METHODS

In various embodiments, the intraluminal physiologic monitoring device comprises one or more sensing subsystems configured to detect physiologic conditions associated with tissue interaction, environmental context, and biologic response.

Bio-impedance sensing may be implemented using single-frequency or multi-frequency impedance spectroscopy techniques. Electrode pairs or electrode arrays positioned along the insertable body may measure electrical impedance characteristics associated with tissue contact, hydration-related proxies, and physiologic variability over time. Multiple electrode configurations may be used including longitudinal arrays, circumferential rings, segmented electrodes, or distributed contact zones.

Capacitive or dielectric sensing elements may be employed to infer moisture conditions, dielectric variability, and tissue interaction properties. Such sensing may utilize conductive pads, embedded electrodes, flexible printed circuits, or conductive polymer structures integrated within the device body.

Pressure and compliance sensing may be achieved using force sensing resistors (FSR), piezoresistive sensors, capacitive pressure elements, strain gauges, or deformable sensing structures. Distributed pressure sensing may enable measurement of localized tissue interaction, compliance behavior, positional stability, and dynamic physiologic response during measurement sessions.

Temperature sensing elements including thermistors, semiconductor temperature sensors, infrared sensing components, or equivalent thermal detection systems may provide physiologic context, environmental compensation, and longitudinal thermal trend analysis.

Optional chemical sensing embodiments may include solid-state pH sensors, ion-selective electrodes, electrochemical sensors, or proxy chemical detection technologies configured to detect physiologic environmental conditions.

Motion and orientation sensing may be implemented using inertial measurement units (IMU), accelerometers, gyroscopes, magnetometers, or multi-axis motion sensing assemblies. Motion data may be used for posture classification, insertion state detection, activity segmentation, and motion artifact rejection.

Certain embodiments may additionally incorporate acoustic sensing, vibration detection, or mechanical response measurement systems configured to capture dynamic tissue-device interactions or physiologic response patterns.

Sensor fusion techniques may combine outputs from multiple sensing modalities to derive higher-order physiologic metrics including contact stability indices, hydration proxy trends, compliance indicators, variability signatures, and longitudinal physiologic profiles.

Sensor configurations described herein are illustrative and non-limiting. Equivalent sensing technologies, future sensor architectures, and alternative measurement approaches capable of achieving substantially similar physiologic characterization are considered within the scope of the disclosure.